Solutions by **Andrew Lys** `alys(at)ncsu.edu`

Problem 1. The stationary distribution with minibatching.

Consider batched SGD defined by:

$$\Phi- = \eta\hat{g}^B$$

Where $\hat{g}^B$ is the average of B sampled gradients. Let $g = \mathbb{E}[\hat{g}]$ The covariance matrix at batch size B is given by:

$$\Sigma^B = \mathbb{E}\left[(\hat{g}^B - g)(\hat{g}^{B\top} - g^\top)\right]$$

The sde model model of SGD is

$$\Phi(t+\Delta t) = \Phi(t) - g\Delta t + \epsilon\sqrt{\Delta t}, \quad \epsilon \sim \mathcal{N}(0, \eta\Sigma^B)$$

Show that for $\eta = B\eta_0$ the stationary distribution is determined by η_0 independent of B.

Proof. We have the following SDE:

$$d\Phi_t = -g(\Phi)dt + \sqrt{\eta}(\Sigma^B)^{1/2}dB_t$$

Let $\mathcal{A}$ be the infinitessimal generator of this SDE. We have:

$$\mathcal{A}^* f = \nabla\cdot(gf) + \frac{1}{2}\nabla^2\cdot(f\eta\Sigma^B)$$

By forward Kolmogorov, we have:

$$\partial_t p = \mathcal{A}^* p$$

To find the stationary distribution, we set $\partial_t p = 0$ and solve the PDE:

$$\nabla\cdot(gp) + \frac{1}{2}\nabla^2\cdot(p\eta\Sigma^B) = 0$$

Let's investigate Σ^B. We have:

$$\begin{aligned}
\Sigma^B[i,j] &= \mathbb{E}\left[(\hat{g}^B[i] - g[i])(\hat{g}^B[j] - g[j])\right] \\
&= \mathbb{C}\mathrm{ov}\left(\hat{g}^B[i], \hat{g}^B[j]\right) \\
&= \mathbb{C}\mathrm{ov}\left(\frac{1}{B}\sum_{b=1}^{B} g[b,i], \frac{1}{B}\sum_{b=1}^{B} g[b,j]\right) \\
&= \frac{1}{B^2}\sum_{b=1}^{B}\sum_{b'=1}^{B}\mathbb{C}\mathrm{ov}\left(g[b,i], g[b',j]\right) \\
&= \frac{1}{B^2}\sum_{b=1}^{B}\mathbb{C}\mathrm{ov}\left(g[b,i], g[b,j]\right)
\end{aligned}$$

Where the last line follows because the samples are independent. Therefore, we have:

$$\Sigma^B[i,j] = \frac{1}{B}\left(\frac{1}{B}\sum_{b=1}^{B}\mathbb{C}\mathrm{ov}\left(g[b,i], g[b,j]\right)\right)$$

Let $\sigma_{i,j} = \mathbb{C}\mathrm{ov}\left(g[i], g[j]\right)$. Since the samples are i.i.d. we have $\mathbb{C}\mathrm{ov}\left(g[b,i], g[b,j]\right) = \sigma_{i,j}$ for all b. Therefore, we have:

$$\Sigma^B[i,j] = \frac{1}{B}\sigma_{i,j}$$

Thus, for all η_0, we have:

$$\eta\Sigma^B[i,j] = B\eta_0\frac{1}{B}\sigma_{i,j} = \eta_0\sigma_{i,j}$$

Which has no dependence on B. Thus, in the above PDE, the solution has no dependence on B. $\square$

Problem 2. The stationary distribution of weighted average updates.

Let $\hat{g}_1, \ldots \hat{g}_N$. Be loss gradients of N individual training points with batch size 1. Consider

$$\Delta\Phi = \sum_i \alpha_i \hat{g}_i$$

Assume the updates are so small so taht for any given training point i we have that $\hat{g}_i$ is unaffected by the drift in the parameter vector. In this case, even if the parameter updates are being made between gradient measurements, the random variables $\hat{g}_i$ are essentially iid. Let Σ_g be the covariance matrix of the distribution of the random variable $\hat{g}_i$. Let $\Sigma_{\Delta\Phi}$ be the covariance matrix of $\Delta\Phi$. Show

$$\Sigma_{\Delta\Phi} = \left(\sum_i \alpha_i^2\right) \Sigma_g$$

Proof. We have:

$$\begin{aligned}
\Sigma_{\Delta\Phi}[i,j] &= \mathbb{C}\mathrm{ov}\left(\sum_k \alpha_k \hat{g}_k[i], \sum_k \alpha_k \hat{g}_k[j]\right) \\
&= \sum_k \sum_{k'} \alpha_k \alpha_{k'} \, \mathbb{C}\mathrm{ov}\left(\hat{g}_k[i], \hat{g}_{k'}[j]\right) \\
&= \sum_k \alpha_k^2 \, \mathbb{C}\mathrm{ov}\left(\hat{g}_k[i], \hat{g}_k[j]\right) \\
&= \left(\sum_k \alpha_k^2\right) \Sigma_g[i,j]
\end{aligned}$$

Since $\hat{g}_k$ are i.i.d. □